

รายงานโครงงาน

ระบบ AI สำหรับ OCR และ Web Scraping เพื่อวิเคราะห์ข้อมูลเชิงลึกทางธุรกิจ

(AI-Driven OCR and Web Scraping System for Business Insight Analysis)

จัดทำโดย

นายจักรภัทร พนมวัฒนศักดิ์ รหัสนักศึกษา 66070026

นายวชิรวิทย์ ธนีสันต์ รหัสนักศึกษา 66070176

อาจารย์ที่ปรึกษา

ดร.ณัฏฐ์ ดิลกธนากุล

รายงานนี้เป็นส่วนหนึ่งของการศึกษารายวิชาโครงงาน

คณะเทคโนโลยีสารสนเทศ สาขาเทคโนโลยีปัญญาประดิษฐ์

สถาบันเทคโนโลยีพระจอมเกล้าเจ้าคุณทหารลาดกระบัง

ภาคการศึกษาที่ 2 ปีการศึกษา 2568

Abstract

This project implements a document assistant for Thai business reports using OCR, table extraction, locally stored embeddings and structured numerical access. We compared a one-pass dense retrieval system (S0) with the selected full application (S1) on the same newly ingested reports, questions, final generation model, context budget and validators. Four 2025 issuer reports contained 1,447 physical PDF pages; 1,431 pages were indexed, with 16 failed checkpoints and additional table-quality warnings retained. Each system produced first outcomes for 498 answerable questions and 32 unsupported controls.

Strict-complete results are 79/498 (15.86%) for S0 and 60/498 (12.05%) for S1. Required numerical matches are 119/356 (33.43%) and 106/356 (29.78%). Appropriate control abstentions remain 24/32 and 28/32. Median response times remain 7.19 and 13.01 seconds. S1 did not outperform S0 on these strict criteria; it abstained more appropriately on controls and took longer.

All 1,060 retained first outcomes now have usable AI-assisted assessments. A separate completion overlay resolves the 348 earlier failed judgments using saved historical responses where available and current Codex review for missing decisions. No new Gemini calls or application answers were generated. Thirty-four assessment records received disclosed consistency adjudications. Original files and the earlier result manifest remain intact; 2,124 frozen evidence hashes were verified. Reviewer assignment was not randomized and no independent human confirmation is available. Three confirmed source-label defects are disclosed with a separate paired sensitivity analysis. The greater-than-80-percent quality goal remains unmet. Costs include known partial components and explicitly unknown charges.

Keywords: Thai business reports; OCR; retrieval augmented generation; numerical context; source provenance; AI-assisted evaluation.

Acknowledgements

The project advisor and institutional information follow the supplied examination report. We acknowledge the original issuer reports as the evidence sources. AI assistance in document extraction, answer generation, evaluation and preparation of this report is disclosed in Chapter 3; it does not constitute independent human verification.

Contents

Abstract 2
Acknowledgements 2
Chapter 1 Introduction 6
1.1 Problem and intended workflow 6
1.2 Objectives 6
1.3 Scope and boundaries 6
1.4 Research questions 6
Chapter 2 Background and Related Work 7
2.1 Retrieval augmented generation 7
2.2 Reasoning and tool use 7
2.3 Official product survey 7
2.4 Numerical provenance 7
Chapter 3 System and Experimental Method 8
3.1 Implemented processing architecture 8
3.2 Matched systems 8
3.3 Inspection workflow 8
3.4 Reports questions and freeze 9
3.5 Frozen first outcomes 9
3.6 Data separation 9
3.7 Evaluation definitions 10
3.8 Paired uncertainty 10
3.9 Execution recovery and AI reviewers 11
Chapter 4 Results and Evidence 12
4.2 Answer and retrieval metrics 13
4.3 Results by category and report 14
4.4 Source QA design 14
4.5 Numerical and citation audit findings 15
4.6 Frozen source labels and paired sensitivity 15
4.7 Traceable cash example 16
4.8 Ingestion reliability and actual tool activity 17
4.9 Retained failures 17
4.10 Operating cost and unknown costs 18

4.11 Illustrative monthly workload 19
Chapter 5 Conclusions and Future Work 20
5.1 Findings 20
5.2 Objectives and practical use 20
5.3 Limitations 20
5.4 Next research steps 20
5.5 Reporting commitment 20
References 21
Author Profiles 22
สมาชิกคนที่ 1 22
สมาชิกคนที่ 2 22
Appendix A Supplementary Source Spot Checks 23
Appendix B Reproducibility and Evidence Index 24

List of Tables

Table 3.1 Locked source reports and final answerable questions 9
Table 3.2 Custom question-level metric definitions 10
Table 3.3 Reviewer provenance and assessment availability 11
Table 4.1 Completed primary results and response time 12
Table 4.2 Assessed means and conservative means for eight custom metrics 13
Table 4.3 Main question categories and retained strict-complete counts 14
Table 4.4 Issuer report results and judgment missingness 14
Table 4.5 Paired source-validity sensitivity excluding three defective questions 15
Table 4.6 Captured tool actions across 530 outcomes per system 17
Table 4.7 Filtered operation attempts and known Gemini components 18
Table 4.8 Illustrative monthly known-token components in USD 19
Table 4.9 Official subscription price examples for desk comparison 19
Table A.1 Stratified source samples and observations 23
Table B.1 Original PDF identities 24

List of Figures

Figure 3.1 Shared processing workflow and separated evaluation evidence 8
Figure 4.1 Recorded primary rates with separate denominators 13

Chapter 1 Introduction

1.1 Problem and intended workflow

Business-report research requires more than locating a number. An analyst must identify the entity, year, unit, consolidation scope and meaning of the figure, then return to the original page. Thai reports combine narrative sections with dense tables and financial notes. A plausible answer can still be unsuitable when it uses a liability instead of an asset, a target instead of an actual result, or a figure from another year. The prototype is intended for a single operator or small research team that repeatedly consults company reports and retains source evidence for later inspection.

1.2 Objectives

The project aims to collect inspectable document sources, extract searchable prose and structured table cells, retrieve numerical and narrative evidence, and generate answers with captured claims, numerical tuples and physical PDF citations. The evaluation objective is to compare the selected full application with a simpler dense baseline under common inputs. A quality target above 80 percent was an objective, not a result; the final measured run did not meet it.

1.3 Scope and boundaries

The official project title retains Web Scraping. In this final measured study, the evidence corpus consists of four downloaded official PDF reports, not a benchmark of arbitrary website extraction. Storage, SQL access and bge-m3 embeddings run locally; historical OCR and generation use online Typhoon and Gemini services. The workflow is therefore not fully offline. Knowledge-graph and Tavily functionality were not required in the selected closure build. No claim of enterprise multi-user readiness, commercial superiority or autonomous financial decision-making is established.

1.4 Research questions

We ask whether the complete application improves strict completion and numerical-relation matches over one-pass dense retrieval; whether it handles unsupported controls appropriately; and what additional response time, tool activity and known-token components it incurs. Source spot checks separately ask whether facts, captured annotations and actual citations agree with the original PDF. These checks are diagnostic and do not retroactively tune the frozen experiment.

Chapter 2 Background and Related Work

2.1 Retrieval augmented generation

Retrieval augmented generation combines a language generator with retrieved external evidence. Lewis et al. [1] studied dense retrieval with external memory for knowledge-intensive tasks. The prototype follows this broad separation: searchable report content supplies document-specific evidence to a final generator. Retrieval does not itself guarantee correctness; relevant passages can be omitted and extracted table headings can be incomplete.

2.2 Reasoning and tool use

ReAct [2] combines reasoning and actions that obtain information from external tools. Here, the full application routes requests to SQL or text retrieval and may attempt search repair. The final study compares entire systems, not an isolated lexical-ranking or SQL ablation. Tool registration is not evidence of use; Chapter 4 reports captured actions. The shared generator and common validators reduce some differences but do not make the systems identical in retrieval behaviour.

2.3 Official product survey

Official product documentation shows that commercial assistants can accept documents and project sources. ChatGPT Plus provides file upload and analysis [6]; Claude Pro includes document-oriented chats and projects [7]; Google AI Pro bundles Gemini services [8]; Gemini Notebook supports PDF and other source formats [9]. This desk survey establishes available features only. No commercial assistant was run on this question bank, so relative accuracy, usability and savings remain untested. The project contribution is its specific Thai numerical context and inspectable experiment records, rather than exclusive access to specialized documents.

2.4 Numerical provenance

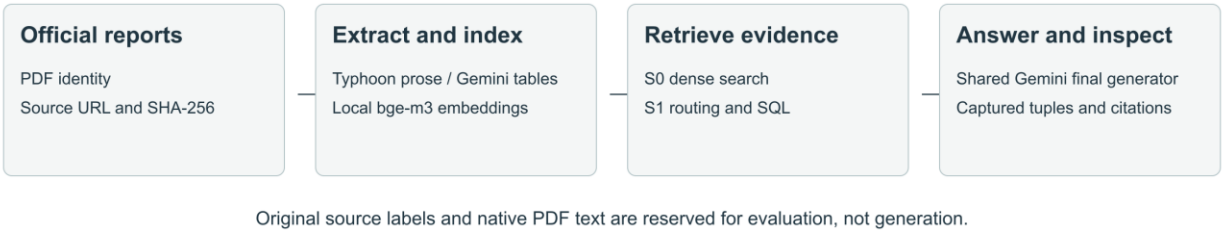
A numerical relation includes entity, measure, value, unit/scale, year, qualifier and source page. A subtraction also needs correctly bound operands and a direction. Physical PDF numbering is used throughout this report; it can differ from printed folios. Original issuer reports [10-13] are used to audit extraction and citations. Human adjudication remains necessary where rounding precision, date duration or reference breadth is ambiguous.

Chapter 3 System and Experimental Method

3.1 Implemented processing architecture

Official PDFs are retained with source URLs and SHA-256 identifiers. The main ingestion route uses Typhoon for page prose and Gemini for table extraction. PostgreSQL with pgvector stores searchable content and page checkpoints; DuckDB stores extracted table cells. Local bge-m3 embeddings have 1,024 dimensions. The final eligible corpus contains 6,928 chunks. These records remain machine derived: an indexed page can still contain table or semantic extraction errors.

Figure 3.1 Shared processing workflow and separated evaluation evidence



3.2 Matched systems

S0 performs one-pass dense retrieval of ten distinct pages with common page expansion. S1 uses the selected application with routing, SQL, lexical/vector retrieval, search repair and deterministic cell handling. Graph-catalog functionality is disabled. Both systems use the same actual OCR corpus, Gemini 2.5 Flash final generator, final-answer prompt, 32,000-character context budget and validators. Source labels and native PDF text are excluded from generation and used only for source evaluation. This is a whole-system comparison; results cannot identify which individual tool caused a difference.

3.3 Inspection workflow

Answers retain text, captured claims, numeric facts, source indices, claim-to-source links and tool traces. Source inspection begins from the emitted document and physical page, checks the original table or paragraph, and confirms the requested entity and numerical dimensions. A truthful citation to a different measure does not make an answer relevant. The live status viewer reads checkpoints only and makes no model requests.

3.4 Reports questions and freeze

Table 3.1 Locked source reports and final answerable questions

Issuer	Physical pages	Indexed pages	Questions
KCE	329	329	125
OSP	375	371	124
HMPRO	470	462	125
TISCO Bank	273	269	124
Total	1447	1431	498

The final answerable bank contains 199 numerical lookups, 100 comparison/multiple-fact questions and 199 narrative/policy/ESG/governance questions. Earlier planned category totals differ from these final locked counts. Each report also has eight unsupported controls, yielding 32 controls per system. The total is 530 outcomes per system and 1,060 across both. Shared question IDs permit paired analysis; two systems answering the same question are not independent question samples.

Four issuer reports provide broad page coverage but remain a small document sample. They were selected as new final reports rather than reused development outcomes. Source references and numerical requirements were prepared with AI assistance and locked before final generation. This is held-out application measurement under that process, not a human-established benchmark. Earlier development scores are excluded from the final tables and are not relabelled as unseen results.

3.5 Frozen first outcomes

The source bank, labels, application code, configuration, prompt, scorer and analysis were frozen for the run. First outcomes, capture errors and refusals were retained. No completed wrong answer was retried to select a passing answer. Operational scheduling and recovery wrappers were disclosed separately. Original source documents and frozen experiment files remain preserved; this report and supplementary QA introduce no database or scorer edits.

3.6 Data separation

Native source text and page images may resolve evaluation ambiguity but must not replace the actual OCR generation corpus. The frozen source-label SHA-256 begins 5959f4e2e4cc. File hashes and record inventories in the appendices support traceability. Hash agreement proves the referenced bytes remained the same; it does not prove that a reference claim or AI judgment is correct.

3.7 Evaluation definitions

Strict-complete success is deliberately demanding. A non-abstaining, error-free captured answer must cover all minimal reference claims, have supported/relevant factual claims, match all required numerical relations, leave no unannotated assertions, and provide at least one fully supporting actual citation per claim. Its prose must explicitly state physical PDF pages consistent with the supported citation pages. A correct narrative with structured citations but no page wording can therefore fail. This metric should not be described as ordinary accuracy.

Table 3.2 Custom question-level metric definitions

Metric	Definition
Faithfulness	Fraction of captured answer claims supported by actual generation context
Factual precision	Fraction of answer claims supported by the original source
Factual recall	Fraction of minimal required source claims covered by the answer
Factual F1	Harmonic mean of question-level factual precision and recall
Claim relevance precision	Fraction of captured claims relevant to the question
Answer relevancy rubric	Judgment score 0, 1 or 2 divided by 2
Context precision AP	Average precision of useful ordered context blocks
Context recall	Fraction of required claims covered by actual context

These are custom source-judgment metrics, not scores produced by the Ragas library. The report shows assessable means with their denominators and conservative means over all 498 main questions. Missing or non-applicable question values contribute zero to conservative means. They are not silently dropped from whole-set results. Claim precision is a mean of question-level fractions, not a claim-weighted micro-average. Numerical rates use 356 required relations per system; controls use their separate 32-question denominator.

3.8 Paired uncertainty

S1-minus-S0 differences use matched main questions. The frozen analysis resamples four issuer-report clusters with 2,000 bootstrap replicates and seed 20261009. This preserves report grouping better than treating every question as independent, but four clusters provide limited generalization. Missingness, reviewer differences and detected judgment inconsistencies limit interpretation of confidence intervals. The intervals do not establish causal superiority or correct an unreliable judgment.

3.9 Execution recovery and AI reviewers

Two unexpected host shutdowns occurred during processing: one during OCR and one after 643 saved generation outcomes. Durable checkpoints and outcome inventories allowed continuation without replaying completed outcomes. The original runner had a resume-only Python scope error. Its source file stayed unchanged; a disclosed wrapper inserted only a global freeze declaration, and removing the declaration restored the original AST. This repaired runtime name lookup without changing questions, corpus, generation or scorer logic.

The failed original resume had already opened DuckDB through existing schema initialization, changing its physical storage hash. A comparison against an untouched warehouse copy including its original WAL verified identical table columns, row counts, canonical row-content hashes and view definitions after the same initialization. The earlier comparison without WAL was incomplete and excluded. Frozen vectors and PostgreSQL corpus identifiers were also checked. Physical byte identity and logical equivalence are distinct claims.

After generation, source-judgment scheduling increased from a sequential worker to disjoint parallel shards, eventually five at the user's request. Prompts, parameters and validation stayed frozen; existing judgments and SDK attempts remained retained. A coordinator later stopped with a OneDrive permission error while owned workers continued. The user then explicitly stopped Gemini. All owned judge processes were stopped, including two in-flight requests whose final provider status and usage remain unknown. No further Gemini calls were made for Codex review or artifact preparation.

Table 3.3 Reviewer provenance and assessment availability

Records	Count	Interpretation
Previously usable assessments	712	604 historical Gemini; 108 Codex
New completion overlays	348	Saved components plus Codex decisions
Consistency adjudications	34	Separate layer; originals preserved
All usable assessments	1060	0 unassessed; 0 human-confirmed

The completed edition retains 712 previously usable assessments and adds 348 separate completion overlays. Existing saved provider responses supplied recoverable components; Codex evaluated missing decisions against the original evidence and frozen validator. A subsequent audit applies 34 versioned consistency adjudications, including prior positive numerical decisions that violated exact quantity, sign, scope or captured-annotation requirements. No original answer or assessment is overwritten. Reviewer assignment was not randomized or blind; source documents and question identities were visible. All judgments remain AI-assisted, including visually inspected source pages. Schema validity does not establish semantic correctness.

Chapter 4 Results and Evidence

All 1,060 first outcomes are retained and all 1,060 assessment records are usable. There are zero unassessed judgments. Tables below come from RESULT_MANIFEST_v 2 using the unchanged scorer and frozen primary references. A canonical input projection changes 16 null annotation collection fields to empty lists without inventing claims, tuples or context. All original outcome files remain byte-identical. Completed assessment coverage is distinct from answer success and from metric applicability.

Table 4.1 Completed primary results and response time

Measure	S0 Dense RAG	S1 Full App
Strict complete /498	79 /498 (15.86%)	60 /498 (12.05%)
Numerical relations /356	119 /356 (33.43%)	106 /356 (29.78%)
False refusals /498	158	176
Capture-error questions /498	34	32
Model-error outcomes	0	0
Unassessed main judgments	0	0
Appropriate control abstention /32	24 /32 (75.00%)	28 /32 (87.50%)
Truthful irrelevant controls /32	8	4
Unsupported requested control assertions	0	0
Observed wall time p 50	7.19 s	13.01 s
Observed wall time p 95	19.90 s	35.55 s

Strict completion differs by -3.82 percentage points for S1 relative to S0. The four-report bootstrap interval is -6.64 to -1.20 points. The numerical rate differs by -3.65 points. More SQL activity did not establish more successful numerical relations. These are paired observations from four reports, with AI-assisted judgments and limited generalization; they do not establish a causal explanation.

Controls ask for unsupported constructed entities or future actuals. The remaining control answers substituted truthful report-supported historical figures that did not answer the request. They are counted as irrelevant substitutions, separately from unsupported assertions about the requested entity/year. Better abstention on controls does not offset false refusals on answerable questions.

4.2 Answer and retrieval metrics

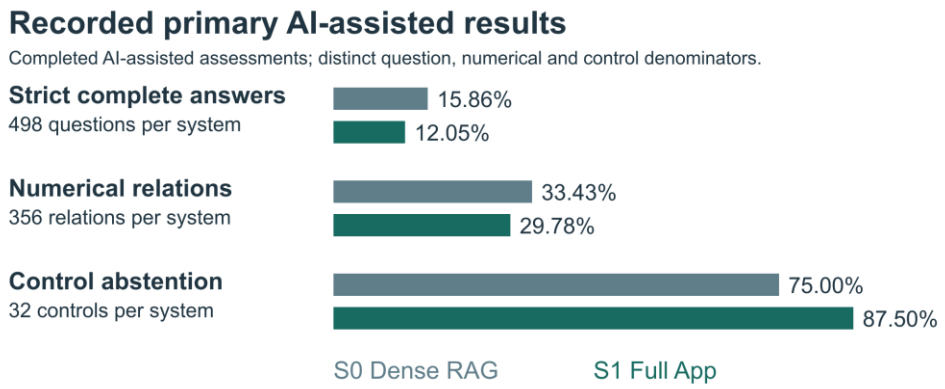
Table 4.2 Assessed means and conservative means for eight custom metrics

Metric	S0 assessed mean / N	S1 assessed mean / N	S0 all 498	S1 all 498
Faithfulness	74.63% / 336	65.81% / 318	50.35%	42.03%
Factual precision	96.38% / 336	92.85% / 318	65.03%	59.29%
Factual recall	55.09% / 498	58.35% / 498	55.09%	58.35%
Factual F1	75.96% / 336	79.89% / 318	51.25%	51.01%
Claim relevance precision	99.35% / 336	99.06% / 318	67.03%	63.25%
Answer relevancy rubric	73.80% / 498	74.80% / 498	73.80%	74.80%
Context precision AP	47.10% / 498	53.39% / 488	47.10%	52.32%
Context recall	38.05% / 498	41.12% / 498	38.05%	41.12%

Assessed-only factual precision is 96.38% for S0 and 92.85% for S1, based on 336 and 318 applicable question values. These are not whole-bank accuracy rates. Refusals and uncaptured claims can have no claim-precision value even though their assessment is complete. The conservative columns retain all 498 questions and assign zero to missing or non-applicable metric values.

Conservative context precision AP is 47.10% versus 52.32%; context recall is 38.05% versus 41.12%. Paired report-cluster intervals are retained in the result manifest. Retrieval averages do not guarantee complete answers when extraction, binding, capture or refusal guards fail.

Figure 4.1 Recorded primary rates with separate denominators



4.3 Results by category and report

Table 4.3 Main question categories and retained strict-complete counts

Category	N each	S0 complete	S1 complete	S0 missing	S1 missing
Comparison / multiple facts	100	6/100	9/100	0	0
Narrative / policy / ESG / governance	199	5/199	6/199	0	0
Numerical lookup	199	68/199	45/199	0	0

Table 4.4 Issuer report results and judgment missingness

Report	N each	S0 complete	S1 complete	S0 missing	S1 missing
HMPRO	125	19/125	16/125	0	0
KCE	125	18/125	12/125	0	0
OSP	124	22/124	22/124	0	0
TISCO Bank	124	20/124	10/124	0	0

There are no missing assessment records in this edition. Genuine non-applicable metric values remain visible. Historical reviewer assignment was concentrated by report and category, and the new completion overlay contains both recovered historical components and Codex decisions. Apparent stratum differences can combine system behaviour, source complexity, extraction quality and assessor effects. Both systems retain the same original outcomes and question denominators.

4.4 Source QA design

One original first outcome was selected from each system/report/category stratum, irrespective of success, using seed 20261009. The resulting 24 spot checks inspect selected original source facts, actual captured numerical tuples and emitted citations. They are supplementary observations, not exhaustive regrading of every claim. Page images and notes are retained with original outcome/judgment hashes. Every stratum is represented; the sample does not certify the entire bank.

4.5 Numerical and citation audit findings

The completion audit found inconsistent positive numerical decisions in historical judgments. For example, KCE03q05 S0 reports source-supported 263,887,772 baht against a frozen 263.9-million reference. This finer-precision quantity is not an exact equality to that reference after unit scaling, so the strict numerical flag is corrected to false in a separate adjudication. Source-supported finer precision is not thereby a hallucination. Semantic reference coverage and factual support are separate criteria. No favorable rounding tolerance was introduced after seeing results.

TISCO06q15 S1 originally marked a five-year answer as a numerical match while its reason called it a mismatch against a four-year subtraction. The source gives BE 2564 to December BE 2568 without a start month; exact elapsed duration is unresolved. The versioned adjudication marks exact-duration support insufficient and the required numerical relation a nonmatch. The original contradictory assessment is preserved. Uniform date-only treatment is also documented for the corresponding paired record and the partial-month tenure question.

Numerical capture and factual text can also diverge. TISCO05q10 S0 correctly states the Stage 1 twelve-month ECL policy at cited page 202 but emits no numeric facts. TISCO06q07 S0 correctly states the tenure dates at cited page 174 but emits no start/end tuples. The frozen numerical criterion fails these captured annotations. For OSP04q08 S0, the answer states an increase of one event while tuples contain only operands 2 and 1; the retained judge credits the derivation through operand indices. Annotation and derivation treatment need calibration before a new benchmark.

Measure identity is a separate failure. TISCO03q03 S1 returns 142,967 thousand baht of long-term employee-benefit expense from page 239, whereas the request's broader employee compensation total is 2,569,856,472.52 baht on page 92. HMPRO05q15 S0 gives truthful lease expiry/renewal details from page 72 but omits the requested recognition policy on page 416. Truthful numbers or paragraphs do not by themselves answer the question.

4.6 Frozen source labels and paired sensitivity

Three source-label defects were confirmed against original reports. KCE01q09 compares Thai laminate/prepreg sales with a Taiwan amount that also includes copper foil. OSP08q12 uses two same-year reporting-scope columns as though they were prior/current years. KCE08q02 asks total separate long-term borrowing but its zero reference is only the fixed-rate subgroup; the floating-rate and balance-sheet tables support 238,100 thousand baht. Primary references remain frozen. These questions are excluded symmetrically only in the following diagnostic sensitivity.

Table 4.5 Paired source-validity sensitivity excluding three defective questions

Criterion	S0 Dense RAG	S1 Full App
Strict complete /495	79/495 (15.96%)	59/495 (11.92%)
Numerical relations /353	119/353 (33.71%)	105/353 (29.75%)

These sensitivity results are not a repaired or newly unseen benchmark. They expose known source-label defects without changing primary denominators. Broader ambiguities include reporting basis, requested versus extra reference facts, partial-date duration and exact rounded precision. Correct prose can still fail if the required captured numeric tuple is absent. Explicit signed decreases may be

equivalent to a negative change; a generic payment row is not silently sign-flipped. All adjudications and their reasons remain separately identifiable.

4.7 Traceable cash example

TISCO Bank page 128 displays BE 2568 cash as 818,057 thousand baht; page 266 displays rounded 818 million. The document, entity, year, scale and requested precision must remain attached to the value. The source check confirms these displays, not a new successful application outcome. This example is used consistently in the poster. Committee duties across page breaks similarly require the preceding heading to avoid assigning an executive duty to the risk committee.

4.8 Ingestion reliability and actual tool activity

Final indexing covered 1,431/1,447 physical pages (98.89 percent). This is checkpoint coverage, not OCR accuracy. Failed physical pages were OSP 8,21,277,320; HMPRO 1,174,191,295,297,321,461,468; and TISCO 79,206,228,229. KCE had all 329 pages indexed but retained table-quality warnings. Unresolved rows and partial semantic extraction problems remain in the corpus and were not relabelled as successful human verification.

Table 4.6 Captured tool actions across 530 outcomes per system

Captured action	S0 count	S1 count
multi_hop	0	22
self_correction	0	7
sql_query	0	314
vector_search	530	435

Tool counts are action occurrences, not necessarily distinct questions. S1 used SQL 314 times and vector search 435 times, with seven self-correction and 22 multi-hop actions. The multi-hop action label does not establish knowledge-graph use. S0 used vector search 530 times. More tool attempts also created more provider calls, which are separated from final outcome counts in cost accounting.

Observed latency includes the time associated with each retained outcome. It is distinct from total elapsed project duration and shutdown downtime. Median S1 wall time was about 1.81 times S0; p 95 was about 1.79 times. These observations describe this machine, provider state and workload. They are not guaranteed service-level performance or a universal parallel throughput limit.

4.9 Retained failures

Capture errors occurred in 34 S0 and 32 S1 main questions. No final outcome was classified as a model error by the frozen summary, but this does not imply zero provider attempt errors. The final generation journals contain 2,322 SDK attempts:2,079 successes and 243 errors, including 234 recorded 429 errors. Provider retries already present in the frozen runner remain in the accounting; completed outcomes were not rerun to improve scores.

4.10 Operating cost and unknown costs

Costs are filtered to the four locked PDF hashes for ingestion and the exact final paired run ID for answer generation. Historical calibration calls are excluded from operation estimates. The known Gemini component uses standard uncached input at USD 0.30 per million tokens and output including reported thinking at USD 2.50 per million, checked against official pricing [3] on 11 October 2026. Cache input is cheaper, but actual cache use is unknown. These components are not total invoices or guaranteed lower bounds: cache could reduce input cost while missing usage/thinking could increase totals.

Table 4.7 Filtered operation attempts and known Gemini components

Operation	Provider attempts	Known Gemini USD	Key unknowns
Four-report ingestion	2,542 total 1,065 Gemini; 1,477 Typhoon	3.436368	64 Gemini usage gaps; thinking unreported
S0 queries /530 outcomes	666	4.074021	54 usage / thinking gaps
S1 queries /530 outcomes	1656	5.610250	189 usage gaps; 934 thinking gaps

Typhoon's official hosted model page advertises a free access tier [4], but the actual account tier and invoice are not established here. Codex review account allocation, hardware depreciation, electricity, administration, tax and credits are also unknown. Interrupted requests with no usage remain unknown. The retained closure research has a separate known Gemini component of USD 29.541234 at cost checkpointv 7; this includes source work and earlier research, and is excluded from monthly operation scenarios.

The accounting retains every SDK attempt rather than counting only successful final answers. Ingestion attempts include reprocessing and provider failures already recorded during the original run. Parallel source review changes waiting time but does not make the evaluation cost disappear. The historical research subtotal is separate from answering on a prepared corpus. Account-level charges cannot be allocated to this project without billing evidence; no numerical value is assigned to unreported Codex usage or local energy.

4.11 Illustrative monthly workload

For a prepared corpus, the recorded known component per outcome is USD 0.00768683 for S0 and USD 0.01058538 for S1. A report of similar average length contributes an illustrative ingestion component of USD 0.859092. Using Q monthly questions and N new reports, the arithmetic is Q times the per-outcome component plus N times 0.859092. This assumes similar report length and behaviour, not measured production scaling.

Table 4.8 Illustrative monthly known-token components in USD

Questions Q	New reports N	S0 component	S1 component
50	0	0.38	0.53
200	0	1.54	2.12
1000	0	7.69	10.59
200	1	2.40	2.98
1000	4	11.12	14.02

Subscription products offer broader services and have different limits and account terms. Partial API components cannot establish subscription-equivalent savings, a break-even threshold or return on investment. A fair economic comparison would need real workload, account invoices, maintenance time and a measured quality/utility comparison.

The source reports average 361.75 physical pages. Reports with more tables, failed OCR pages or longer retrieved contexts may consume more attempts and tokens than this arithmetic assumes. Cache use and provider pricing can also change. The monthly examples are planning scenarios based on the recorded component, not promises of a production bill. Repeated queries were not rerun here to measure cache savings or concurrent response capacity.

Table 4.9 Official subscription price examples for desk comparison

Product	Listed monthly USD	Comparison scope
ChatGPT Plus [6]	20.00	File analysis; API billed separately
Claude Pro [7]	20.00	Monthly subscription; annual option differs
Google AI Pro [8]	19.99	US monthly listing; broader Google bundle

These official listings were checked on 11 October 2026. They do not establish Thai checkout prices, taxes, product quotas for this workload or comparative answer quality. No subscription was purchased or tested as part of this final study. The prototype’s inspectable storage and source records are potential workflow benefits, but a useful comparison must include the time spent checking unreliable answers.

Chapter 5 Conclusions and Future Work

5.1 Findings

The project produced an inspectable Thai business-report processing and answer workflow and completed the paired first-outcome run. The selected full application did not outperform one-pass dense RAG on the preserved strict-complete and numerical criteria. It handled unsupported controls more appropriately but increased median and tail response times. Slightly higher retrieval averages did not resolve answer binding, refusal or annotation problems. The greater-than-80-percent quality goal remains unmet.

5.2 Objectives and practical use

Document collection, durable page processing, local evidence storage, numerical access and source inspection were demonstrated. Automatic answer reliability was not demonstrated at the target level. The prototype can support supervised evidence discovery, with the operator opening source pages and checking entity, period and units. It should not present generated figures as verified business conclusions merely because a document citation is attached.

5.3 Limitations

Only four reports were evaluated. Sixteen failed page checkpoints and table warnings restrict the corpus. All assessments are complete, but source labels and judgments remain AI-assisted and nonrandomly assigned. Three confirmed source-label defects and other scope, precision and duration ambiguities are disclosed. Strict page-wording and captured-tuple requirements can reject factually supported text. Unknown costs prevent a complete economic comparison.

5.4 Next research steps

A new evaluation should begin with independent human adjudication of reference scope, rounding tolerances and duration definitions, followed by calibrated semantic judgment checks. Retrieval and capture issues should be corrected in a new explicitly versioned build, then compared with the baseline on a newly frozen held-out set. Failed OCR pages and ambiguous table headers need systematic repair and verification. Any later commercial comparison should run the same permissible document tasks under documented product settings and report uncertainty. These proposals are future work, not completed improvements or permission to retune this frozen run.

5.5 Reporting commitment

All primary values in this report and poster come from the completed versioned result manifest. Original first outcomes, original judgments and the v 1 manifest remain preserved. Completion overlays and 34 assessment adjudications are distinct from system improvements. Paired source-validity sensitivity does not replace frozen primary results. Original project, member, advisor, institution and academic-term identities remain unchanged.

References

[1] Lewis, P., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. https://arxiv.org/abs/2005.11401

[2] Yao, S., et al. (2023). ReAct Synergizing Reasoning and Acting in Language Models. https://arxiv.org/abs/2210.03629

[3] Google Cloud. Gemini model pricing. Official standard Gemini 2.5 Flash input/output rates, verified 11 October 2026. https://cloud.google.com/gemini-enterprise-agent-platform/generative-ai/pricing

[4] Typhoon. Typhoon OCR hosted model and access information. https://opentyphoon.ai/model/typhoon-ocr

[5] Google Cloud. Throughput and quota for generative AI. https://docs.cloud.google.com/vertex-ai/generative-ai/docs/resources/throughput-quota

[6] OpenAI. What is ChatGPT Plus. https://help.openai.com/en/articles/6950777-what-is-chatgpt-plus

[7] Anthropic. Claude pricing and Pro features. https://claude.com/pricing

[8] Google. Google AI plans. https://one.google.com/about/google-ai-plans/?hl=en&gl=US

[9] Google. Gemini Notebook supported source formats. https://support.google.com/gemininotebook/answer/16215270?hl=en

[10] KCE. One Report 2025 Thai edition. Original retained PDF, 329 physical pages. https://kce.listedcompany.com/misc/onereport/20260323-kce-onereport2025-th.pdf

[11] OSP. One Report 2025 Thai edition. Original retained PDF, 375 physical pages. https://hub.optiwise.io/storage/35/annual-report/2025/osp-one-report2025-th.pdf

[12] HMPRO. One Report 2025 Thai edition. Original retained PDF, 470 physical pages. https://hmpro.listedcompany.com/misc/one-report/20260309-hmpro-one-report-2025-th.pdf

[13] TISCO BANK. One Report 2025 Thai edition. Original retained PDF, 273 physical pages. https://www.tisco.co.th/wp-content/uploads/2025-bank-annual-report-th.pdf

[14] Project frozen first-outcome records, source labels, RESULT_MANIFEST_v 2, filtered cost components and supplementary QA. Local evidence index in Appendix B. AI-assisted records; not independent human validation.

Author Profiles

สมาชิกคนที่ 1

ชื่อ-สกุล: นายจักรภัทร พนมวัฒนศักดิ์

รหัสนักศึกษา: 66070026

สาขาวิชา: เทคโนโลยีปัญญาประดิษฐ์

คณะ: เทคโนโลยีสารสนเทศ

สถาบัน: สถาบันเทคโนโลยีพระจอมเกล้าเจ้าคุณทหารลาดกระบัง

อีเมล: 66070026@kmitl.ac.th

สมาชิกคนที่ 2

ชื่อ-สกุล: นายวชิรวิทย์ ธนีสันต์

รหัสนักศึกษา: 66070176

สาขาวิชา: เทคโนโลยีปัญญาประดิษฐ์

คณะ: เทคโนโลยีสารสนเทศ

สถาบัน: สถาบันเทคโนโลยีพระจอมเกล้าเจ้าคุณทหารลาดกระบัง

อีเมล: 66070176@kmitl.ac.th

Appendix A Supplementary Source Spot Checks

The original 24 stratified samples cover both systems, four reports and three categories. Their dated findings were diagnostic observations from the earlier assessment checkpoint. This edition separately completes all missing judgments and applies disclosed consistency corrections; original sample notes remain preserved. The sample does not certify all claims or confer human confirmation.

Table A.1 Stratified source samples and observations

Record ID	Arm	Original pages	Selected observation
F_HMPRO_08__q09	S1	464	Bond difference -1116; refusal
F_HMPRO_03__q10	S1	169,155	Paris claim supported; null units
F_HMPRO_05__q06	S1	206	Contractor LTIFR 0.32; refusal
F_KCE_02__q08	S1	91	Revenue decrease 1871.8; refusal
F_KCE_04__q13	S1	164	Appointment source 164; refusal
F_KCE_01__q06	S1	29	15 suppliers and citation agree
F_OSP_03__q07	S1	126	ROE change 12.39 pp; refusal
F_OSP_07__q15	S1	248,12	Selected teamwork claim supported
F_OSP_08__q01	S1	364	Lease carrying 158366; source supported
F_TISCO_BANK_06__q15	S1	174,96	Duration ambiguity; adjudicated
F_TISCO_BANK_05__q15	S1	205	Revaluation policy/citation agree
F_TISCO_BANK_03__q03	S1	92,239	Employee measure mismatch
F_HMPRO_02__q07	S0	129	Income difference -2006.74; refusal
F_HMPRO_05__q15	S0	416,72	Truthful lease details; wrong policy
F_HMPRO_08__q02	S0	459,313	Rounded cash rejection inconsistency
F_KCE_08__q09	S0	323	Singapore cash -6486; refusal
F_KCE_03__q14	S0	76	Wrong-company retrieval; refusal
F_KCE_03__q05	S0	92,239	Fine precision; strict nonmatch
F_OSP_04__q08	S0	102	Correct operands; derived capture issue
F_OSP_04__q11	S0	96	Selected human-rights claims supported
F_OSP_06__q06	S0	153,157	17 directors; valid alternate page
F_TISCO_BANK_05__q07	S0	85,107,184	7 attendances vs 8 regular meetings
F_TISCO_BANK_05__q10	S0	202	12-month policy; missing tuple
F_TISCO_BANK_06__q07	S0	174	Correct tenure text; missing tuples

Appendix B Reproducibility and Evidence Index

Table B.1 Original PDF identities

Report	SHA-256 prefix	Physical pages
KCE	7f975d7d1a31d3e6cc4c	329
OSP	21e932eba2dbc63dc65e	375
HMPRO	bf6f0db9e41653078bc2	470
TISCO_BANK	bfec42f8da8ca83cd282	273

All paths below are relative to TestFile/project_closure_2026-10-09. RESULT_MANIFEST_v 2 is the sole completed primary summary; v 1 remains a historical checkpoint. Completion provenance identifies each retained assessment, overlay and adjudication. The frozen scorer was replayed with four path constants substituted; numerical and metric algorithms were unchanged. Canonical projection only converts null annotation collections to empty lists.

Primary results: final_analysis_v2/RESULT_MANIFEST_v2.json

Per-question rows: final_analysis_v2/RESULTS_BY_QUESTION.json

Filtered cost components: final_analysis_v1/FILTERED_COST_COMPONENTS_v1.json

Locked source labels: final_bank_source_lock_v1/source_labels530.json

First outcomes: final_paired_v24_v1/first_outcomes/

Original and Codex judgments: final_source_judgments_v1/judgments/

Completed reviewer provenance: evaluation_completion_v2/REPLAY_PROVENANCE.json

Assessment overlays and corrections: evaluation_completion_v2/

Integrity audit: evaluation_completion_v2/INTEGRITY_AND_COVERAGE_AUDIT.json

24 source spot checks: codex_source_review_v1/STRATIFIED_QA_FINDINGS.json

Original 952 hash inventory: codex_source_review_v1/ORIGINAL_952_JUDGMENTS_SHA256.json

Stop and reviewer change: codex_source_review_v1/GEMINI_STOP_AND_REVIEWER_CHANGE.json

Runtime binding repair: reboot_recovery_20261010_v3/RESUME_BINDING_FIX_DISCLOSURE.json

Warehouse logical proof:
reboot_recovery_20261010_v2/WAREHOUSE_WITH_WAL_STANDARD_INIT_EQUIVALENCE.json

Primary manifest SHA-256: 130bdf6292060a6771cedfef89450ad1f3be87d385d7164fcab53ed77ec1b391

The final Word source, PDF, poster SVG/HTML, print PDF and preview PNG are generated locally without new Gemini calls. Word contents and lists are updated during native rendering. Visual QA checks every rendered report page and the complete A1 poster. This artifact QA checks presentation; it does not change the false quality-goal flag or confer human-confirmed experimental validity.